

COMPONENTS OF CCTV SYSTEMS

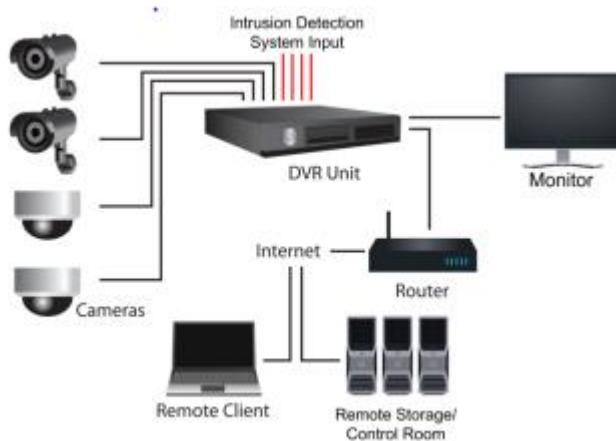
CCTV uses components that are directly connected to generate, transmit, display, and store video data.

A CCTV system can be as simple as a camera purchased from a retail electronics store connected to a video monitor. However, larger systems operated by professional security personnel are comprised of a number of components falling into several basic categories:

- Cameras;
- Lenses;
- Housings and mounts;
- Monitors;
- Switchers and multiplexers; and
- Video recorders.

The most complex CCTV systems may incorporate hundreds of cameras and sensors integrated into one overall security network.

Figure 3-1 provides a CCTV component diagram example.



Cameras

Cameras are an essential component of any CCTV system. Matching the right CCTV camera to a particular application is increasingly complex due to rapid technological developments and a greater range of applications.

All CCTV cameras include three basic elements:

- **Image sensor**—Converts light (photons) into electronic signals;
 - **Lens**—Gathers light reflected from a subject and focuses the light on the image sensor;
- and
- **Image processing circuitry**—Organizes, optimizes, and transmits video signals.

The type of camera best suited for a CCTV system depends on the operational environment and how it will integrate into the system. The answers to the following questions may help determine the best camera type:

- What is the desired image quality?
- What size is the desired field of view (FOV)?
- How much lighting is available?
- Will the camera be installed indoors or outdoors?
- Will the video be monitored on a full time basis?
- How will the video be transmitted?
- Will the camera be exposed to extreme conditions?

Classification of cameras according to

1.Fixed or ptz

There are many types of cameras designed to perform under specific environmental conditions but cameras can be grouped into two primary categories: fixed and pan-tilt-zoom (PTZ).

Fixed cameras are intended to constantly view a single scene, while **PTZ cameras** are motor driven and can pan left or right, tilt up or down, and zoom in or out to instantly customize the view as needed. A combination of fixed and PTZ cameras are often used to provide the required surveillance coverage.

3.1.1 Fixed Cameras

Fixed cameras are mounted in a stationary position and are focused on a single FOV, typically one particular area of interest. These cameras can be used indoors or outdoors and can be installed overtly or covertly.

Fixed cameras are usually less expensive than PTZ cameras and require less maintenance as they have fewer moving parts.

3.1.2 PTZ Cameras

PTZ cameras come in a variety of sizes and shapes for interior and exterior uses. Typically, a

PTZ camera can be turned and tilted on two axes to provide pan and tilt capabilities and the focal length of the lens can be varied to change the FOV. This enables PTZ cameras to offer more flexibility for viewing and capturing images in real time than fixed cameras. PTZ cameras can be operated manually or in an automatic scan mode, thus capturing the most relevant video possible.

2. Connectivity Type

CCTV cameras may employ one of two types of data transmission:

Network Cameras—Network cameras connect to IP-based networks, including the Internet, and provide remote viewing and recording. Network cameras are also available in high definition (HD) which can provide greater image detail.

Analog Cameras—Despite increasing use of digital network cameras, a market for analog Cameras still exists. This may be due to the cost involved in upgrading and converting to a new transmission process. Analog cameras have options for high resolution, making them applicable for various surveillance needs. These cameras also have some cyber security advantages because the coaxial cable they are connected with would require physical access to breach

3. Day/Night Cameras

Day/night cameras offer flexibility by automatically adjusting to current lighting conditions.

These cameras capture color images in daylight and switch to black-and-white to improve image quality at night. The camera relies on an analysis of the current image or a photoelectric sensor to determine when to automatically remove the infrared-cut filter and switch to monochrome settings

4. Low-Light or Night Vision Cameras

Cameras used to capture images in dark environments are either low-light or night vision cameras.

Low-light cameras are designed to perform in some level of ambient lighting, such as indoor restaurant lighting, street lamps, or a full moon; they are not intended for use in complete darkness.

Night vision cameras used in CCTV systems typically consist of near-infrared (NIR) and IR cameras with built-in IR illuminators. They are designed to allow the operator to view night scenes. The distance from which a CCTV camera can detect objects at night depends on the capability of the camera components, such as the lens and sensor, as well as on the intensity of IR illuminator used.

5. Thermal Imaging Cameras

Some operational environments may require a thermal imaging camera to detect through obstructions such as fog or smoke. Thermal imaging cameras detect infrared or heat radiation that is invisible to the human eye. Currently these cameras are sensitive to a temperature difference of one tenth of a degree Fahrenheit. Thermal imagers cannot detect through glass or water, but can provide an image through limited density fog or smoke.

6. Miniature or Covert Cameras

Special applications may exist that require the installation of small, hidden cameras as part of a CCTV system. These cameras are not usually weather resistant, so they may require an external housing if they are to be used outdoors. These cameras are typically battery operated and may use built-in transmitters to provide a compact wireless solution. Organizations can choose from many types of miniature and covert cameras depending on their requirements

Types of Lenses

Lenses are available in three basic types: fixed focal length, varifocal (variable focal length), and zoom. The focal length of a lens is the distance between the optical center of the lens and the image plane. The lens focal length and the image sensor size determine the camera's FOV.

Fixed Focal Length Lenses

Fixed focal length lenses are built with one set, unchangeable, focal length. Such lenses are useful in situations where the camera remains in a fixed position and the requirements to observe an area do not change.

Varifocal Lenses

The focal length of varifocal lenses can be changed within a specific range; however, each change must be accomplished by hand at the camera. In addition, each time the focal length is changed, the iris and focus may also need adjustment. Varifocal lenses offer the flexibility of varying the scene content and are relatively inexpensive when compared to conventional zoom lenses.

Zoom Lenses

Unlike the varifocal lens, the zoom lens is designed to maintain the focus setting throughout the focal length range. Zoom lenses in the CCTV industry are often built with integral motors to enable changing the focal length from a remote location. They are used on cameras that monitor different parts of a scene or can pan and tilt to monitor different locations.

Monitors

This section describes various types of monitors that are available on the market to meet an organization's CCTV

Televisions—HDTVs can be used to view video in certain circumstances such as viewing video for investigation purposes from a secured office. Televisions can also offer a cost-effective alternative for dual purpose monitoring of both internal CCTV feeds and network broadcasts. However, televisions do not have the same durability as monitors; therefore, it is not an advisable choice for 24-hour monitoring requirements.

PC Monitors—DVRs can be connected to a PC monitor. Inputs and outputs between these devices need to be matched using standard connectors, adapters, or converters. **Monochrome Monitors**—Due to the differences in technologies, monochrome monitors have a greater potential to display detail than color monitors. Although the difference in resolution of color and monochrome monitors is narrowing, the potential loss of visual detail with a color monitor must be weighed against the cost and benefits of its display. Either type of monitor can display both monochrome and color video signals.

CCTV Monitors—Monitors designed for CCTV systems are preferred if the system will include multiple monitors and cameras, or if the monitor is always on. Monitor technologies include cathode ray tube (CRT), LCD, organic light-emitting diode (OLED), and plasma.

CRT Monitors—The CRT analog monitor is a legacy CCTV monitoring technology. CRT monitors are available in a variety of sizes. CRT monitors should include electromagnetic interference (EMI) protection. EMI can distort the path of the electron beams and reduce the quality of the image. Since CRT monitors are bulky, they are no longer used in handheld viewing devices and are seldom used in other configurations of portable equipment.

LCD Monitors—LCD monitors are available in many shapes, sizes, and resolutions. They provide a sharper, brighter picture than CRT monitors because they can process CCTV signals at a higher resolution. LCD monitors are much thinner front-to-back than CRT monitors of similar screen size so they take up less space and they generate less heat. Many are small and compact enough to be portable. They are also available in large sizes for multiple camera viewing. An important advantage of LCD monitors over CRT monitors is that LCD monitors do not suffer from magnetic interference.

OLED Monitors—LED/OLED monitors use OLEDs, which emit light directly without the need for a backlight. This means that each diode can be turned on and off depending on what needs to be displayed on the monitor screen, unlike LCD displays, which require the backlight source be covered to produce blacks. The OLED implementation results in better contrast as well as reduced power consumption compared to LCD monitors. However, OLED display technology is more expensive, more difficult to produce, and has a shorter lifetime than LCD monitors.

Plasma Monitors—Plasma monitors are much like LCD monitors in that they are flat panels and individually control every pixel. A plasma display has a specific gas inside every pixel. The pixel elements illuminate in response to received electrical signals. Plasma monitors are available in many different sizes, although most are manufactured in a wide-screen 16:9 aspect ratio. The main advantage of plasma over LCD monitors is the greater viewing capability from an off-axis angle.

Table 3-3. CCTV Monitor Technology Comparisons

Type	Pros	Cons*
CRT	• Good overall picture quality • Robust technology • Low-cost option that is compatible with most existing CCTV equipment	• High power consumption • High heat generation • High space requirements • Manufacturers have largely discontinued • Burn-in tendency (when consistent image is permanently etched onto screen)
LCD	• Compact and relatively light • Low power consumption and electromagnetic radiation • Wide range of screen sizes available • No flicker • Less burn-in. Long life expectancy (up to 50,000 hours)	• Limitations reproducing true black colors • Restricted viewing angle • Low image contrast • Time lag of pixels can lead to “smearing” effect
OLED	• Thinner, lightweight, low-power consumption, and better dynamic contrast ratio • Brighter display and wide color range	• More expensive than other monitor technologies • Shorter lifespan
Plasma	• Wider viewing angles than LCD • Higher black levels • Higher contrast ratio (the difference between the brightest and the darkest part of the image)	• Fragile • High power consumption • High heat generation • Susceptible to burn-in

Types Of Video Recorders

1.Digital Video Recorders

A CCTV system may send digital or analog video to the recording system. A DVR receiving analog video takes two fields of the analog signal and builds one image, which is then digitized and compressed. If the video going to the DVR is digital, it is normally compressed to save storage space. Various data compression methods can be used that offer varying degrees of performance, quality, and storage economy.

2.Hard Drive Recorders

Standard rack-mount, hard drive recorders are generally mounted in a 19-inch rack system and may receive a number of analog video inputs: 4-, 8-, 12-, 16-, and 32-input units are typical.

These units can be switched between the NTSC and phase alternating line (PAL) video standards and have power supplies that accept a 60-Hz or 50-Hz current.

The hard drive capacity is a key specification for systems without a network storage strategy,

since the device hard drive will determine how many days of recording remain before the system begins recording over the oldest video. The data storage requirements will depend on the type of cameras connected to the DVR. The resolution, image rate, motion detection features, complexity of image, and FOV will also affect the unit's storage availability.

Most hard drive recorder models have the following features:

- Multiple hard drives;
- Built-in compact disk rewritable (CD-RW) or digital versatile disk (DVD) that provides permanent recording on removable media;
- Alarm inputs to trigger event recording; and
- Multiple analog monitor outputs, usually up to four.

3.Network Video Recorder—NVRs record digital video data transmitted over an IP network from multiple CCTV digital cameras. NVRs can be configured to record video in a digital format to a disk drive, portable storage device or a mass storage device. NVRs differ from DVRs as video input is delivered from an IP network. With an NVR configuration, video is encoded and processed at the camera and streamed to the NVR for storage or remote viewing. With a DVR configuration, video is encoded and processed at the DVR.

4.Hybrid DVR—Hybrid DVRs incorporate functions of both NVRs and DVRs by accepting and converting video images from analog cameras as well as IP video inputs. Hybrid DVRs often have IP connectivity supporting remote viewing capabilities to play back recorded video from across the network. They also allow viewing of live video.

5.Mobile Recorders—Mobile recorders are replacing mobile VCRs in patrol cars and on motorcycles. Mobile recorders have all the features available in a conventional DVR and are less susceptible to dirt. These recorders have become reliable recording platforms due to the advancements in immunity to vibration and motion. Many mobile recorders provide the ability to download recorded data to a memory card.

Portable Storage Devices

DVRs can accommodate several types of portable storage media. Universal serial bus (USB) flash drives, memory sticks, and DVD+R dual layer (DL) are detailed below.

1.USB Flash Drives—USB flash drives allow video to be recorded or transferred. USB flash drives are available in capacities in excess of 128 GB, have a fast transfer speed, and can be programmed with password protection.

2.Memory Sticks—Memory sticks come in a variety of capacities up to 1 terabyte (TB). They have a fast transfer rate allowing video to be transferred to them quickly.

3.DVD+R DL—DVD+R DL discs have two storage layers and can hold about 17 GB of data.

There are also other DVD media available, including DVD-R, DVD-RW, DVD+R, DVD+RW, and DVD-RAM. These are lower in cost but hold less data (approximately 4.9GBs).

TRANSMISSION

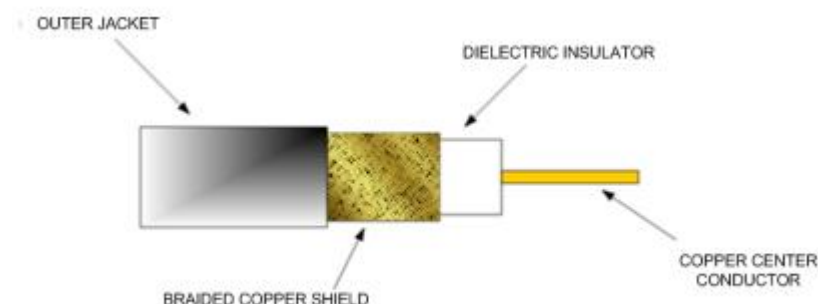
The transmission system is an important component of the CCTV imaging chain that sends and receives video signals between the cameras, the processing system (i.e., DVRs, NVRs, and multiplexers) and the monitoring system (i.e. the display). Transmitting a strong video signal with low noise is vital to producing a high-quality image on the monitor

a) Wired Transmission

Wired CCTV systems use cables to connect cameras to other CCTV components. Wired transmission can provide good quality video images with fewer instances of interference because cables are shielded. Cameras can be located far away from recording or monitoring equipment. Three types of wired CCTV systems are commonly used today: coaxial cables, UTP cables, and fiber optic cables.

1.Coaxial Cable

Coaxial cable is the most common method of transmitting video signals from the camera to the monitor or other CCTV components. Coaxial cable consists of a single wire surrounded first by a nonconductive insulating layer (dielectric), then by a braided wire shield, and finally a plastic or rubber covering.



2.UTP Wire

In some cases, running coaxial from a camera to a monitoring location is not practical and existing telephone wire can be used. For example, many buildings contain abandoned telephone lines, known as UTP, which can be used for a video system. This has several advantages: overall cost savings, low susceptibility to EMI or induction, no ground loop concerns, and ease of use.

Also, telephone wire is smaller and much lighter than coaxial cable. It should be noted that using abandoned telephone wire beyond a facility's boundaries may require an approval process and service agreement with the telephone company.

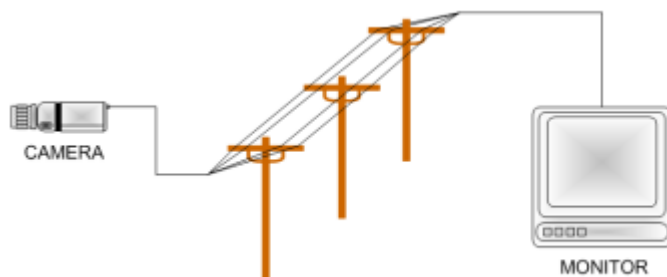
3.Fiber Optics

Fiber optic cable is lightweight and made up of a single spun glass or plastic fiber or a group of such fibers encased in a protective covering. It has a broad bandwidth, making it ideal for carrying video signals. Fiber optic cable can be used in runs up to 6 miles without amplification.

The video signal coming from the camera must first go through a fiber transmitter which converts electrical signals to light impulses. A fiber receiver at the other end is required for conversion back into electrical signals.

4.Telephone Network

Another option for wired transmission of video signals is the telephone network. Although standard voice grade telephone lines do not have enough capacity to handle real-time full motion video, they still have value in specialized CCTV applications. However, telephone lines are not recommended when the security of the video is a concern due to the cyber security vulnerabilities.



5.Category 5 Cable

Networks transmit video over Category 5 cable. The cable consists of four pairs of UTP 24-gauge copper wire with three twists per inch. The high number of twists per inch reduces the “crosstalk,” or EMI, between signals passing on the strands of the cable. Category 5 cables can be used to carry frequencies of up to 100 megahertz (MHz) and handle data rates up to 1,000 megabits per second (Mbps). The cables are terminated with an RJ45 connector and must meet the Electronics Industry Alliance/Technology Industry Association 568 Commercial Building Telecommunications wiring standard.

b) Wireless Transmission

Wireless options for transmitting video can be advantageous due to ease of installation, lack of cabling requirements, and assured mobility. There are some disadvantages of wireless systems such as the need for a dedicated frequency to transmit signals, signal interruptions, and signal interference. There are also increasing concerns over cyber security and the need for information assurance hardening. Wireless transmission options for CCTV systems are discussed below.

1.Laser

In some applications, it is impractical to install video transmission cables. If there is a direct line-of-sight between the transmitting and the receiving locations, laser or IR transmission may be used.

Applications for laser transmission versus wired transmission offer several advantages. Laser transmission systems offer the following features:

- *Transmission over distances of 20 meters to 2 kilometers in clear weather;*
- *Data speeds from 25 MB to 1.2 GB;*
- *Handling of misalignments of 2.4 degrees horizontally and vertically (particularly useful for tall buildings or towers that sway in the wind);*
- *Indoor mounting applications that can be placed behind a glass window;*
- International Electrotechnical Commission Class 1M safe (can be safely viewed by the naked eye);*
- *No ground loop, lightning, or induction issues;*
- *No license or permit requirements; and*
- *Difficult to tap into without being detected.*

Lasers also have disadvantages. Lasers should not be placed in areas with human activity to prevent possible retinal injury. In addition, laser transmission is affected by weather, a factor that influences the feasible distance between the transmitter and the receiver. Line of sight and alignment of the signal is critical. Recent technological advances have included a narrow beam and auto tracking capabilities that tolerates fog and snow better than older laser systems. While there is always the possibility of momentary disruption, retransmission can automatically occur.

2 Infrared

An alternative to laser transmitters is an IR LED transmitter. IR LED systems may cost less than laser systems, but they often have shorter transmission distances and lower bandwidth. LED systems are also more susceptible to reflections when transmitting through glass. IR receivers should not be oriented so that they face the sun, incandescent light bulbs, or other IR light sources. When considering an IR LED transmission system, it is important to clearly understand the application and installation constraints.

3 Radio Frequency

Radio frequency (RF) is another method of wireless transmission relevant to CCTV surveillance. RF modulation of video signals is used in cable TV systems with ultra high frequency (UHF) and very high frequency (VHF) bands. RF encompasses a broad range of frequencies. Transmissions are typically focused beams, which are most efficient in open-air, line-of-sight situations. Physical obstructions such as walls or buildings are likely to degrade an RF video signal.

A typical CCTV RF system may:

- Operate on 2.4 to 5.8 GHz bands;
- Interface with 802.11 access points so that software can be upgraded via a network;
- Require Federal Communications Commission (FCC) licensing;
- Include video compression; and
- Include remote device/sensor control.

RF systems are often used when a cable installation or lease of a high-bandwidth line is impractical. Limited range is a common misperception regarding RF transmission systems. In fact, RF CCTV transmission systems can operate over long distances using directional antennas and repeaters. RF can transmit approximately 2 miles with a flat antenna array or up to 7 miles with a directional antenna.

4.Microwave

Microwave transmission is a wireless, line-of-sight transmission medium with many similarities to laser and LED systems. Depending on its specific configuration, a microwave system can transmit video, audio, and data. Some factors to consider when exploring microwave transmission are:

- Transmissions from 100 feet to 20 miles are possible, and longer distances can be achieved with microwave repeaters;
- Microwave transmitters may require FCC licensing;
- Microwave transmission tolerates adverse weather and obstructions better than laser transmission. However, very heavy rainfall and very dense fog can affect microwave transmissions;
- Microwave systems emit at low energies, which are typically less than 1 milliwatt per square centimeter (mW/cm²). In comparison the minimum safe radiation exposure level established by the American National Standards Institute is 10 mW/cm²; and
- Microwave systems typically are more expensive than LED or laser systems.

c) IP Network Transmission

IP-based systems have emerged as an attractive alternative to other technologies, due in large part to their ability to achieve high-performance video capabilities at a low cost. The industry has found ways to implement IP-based systems that use existing cameras, cables, and other equipment. However, organizations planning and designing new systems should consider IP-based technology. This section addresses the basic parameters of an IP-based network system.

Internet Protocol Network System Overview

IP-based CCTV systems are designed to provide the ability to monitor, record, and stream video over a network to computers or other equipment. The system can use existing local area

networks (LANs), wide area networks (WANs), and/or wireless LANs (WLANs) to save on installation costs. However, for added security, an organization could install its own private area network (PAN) cabling and support hardware. Power over Ethernet (PoE) technology is also an option within an IP-based system to increase savings and reliability. PoE enables various networked devices to receive power and data through one standard cable, which can be a significant cost savings when designing CCTV systems.

A simple IP-based CCTV system, such as the one seen in Figure below consists of a network camera (although analog cameras can be used with additional equipment), a network switch, and a PC for viewing, storing, and analyzing data and managing the CCTV system.

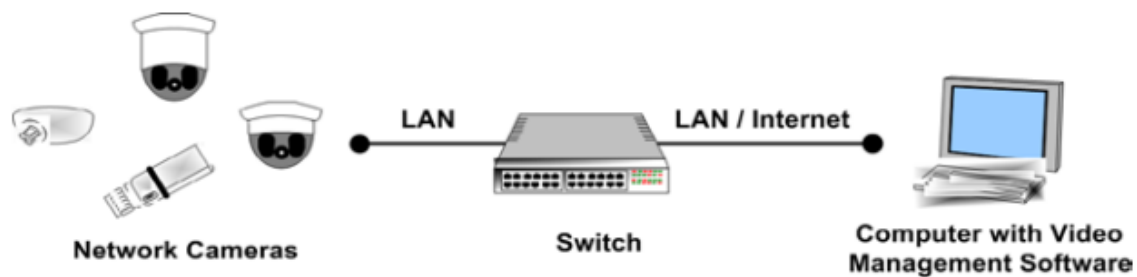


Figure 4-5. IP-Based CCTV System

Benefits of IP-Based Systems

Digital systems in general have a variety of advantages over analog systems such as ease of use, advanced search capabilities, simultaneous record and playback, improved image quality, and efficient compression and storage options. IP-based systems also provide many benefits that include:

- Remote accessibility;
- High image quality;
- Future integration with digital technologies;
- Flexibility;
- Scalability; and
- Cost-effective transmission

IP-Based System Components

The flexibility of IP-based systems is attributed to the variety of configurations and types of components compatible with IP technology. Since the number of possible custom configurations is so vast, the following list is just a sample of the type of components compatible with IP-based systems.

Cameras—Both IP network cameras and analog cameras can be used in an IP-based system.

Video Encoders—When using analog cameras, a video encoder or video server needs to be

connected to the analog cameras to convert the video to a digital format. The encoder then sends the data over an IP network.

Network Switches—Switches allow CCTV devices to communicate with each other and share information.

Networks—A network can be small or extensive, wired or wireless or a combination thereof.

The most common approach taken by organizations is to use LANs or WANs. Network bandwidth capacity can be increased by adding switches and routers. Wireless networks are a good option when traditional wired networks are too costly or difficult to install.

Power over Ethernet—PoE is an option for using a wired network to distribute both data and power.

PC with Web Browser—PCs can access live and recorded video over the Internet as needed. CCTV Technology Handbook

PC with Video Management Software—PCs can record and store video from cameras, as well as view live and recorded video as needed. Additionally, video management software can support video being accessed over smartphones or tablets.

Storage Devices—Video transmitted through an IP system can be stored on a server, a network device such as direct attached storage (DAS), storage area networks (SAN), network attached storage (NAS), or a PC hard disk.

Mobile Devices—IP-based systems can be easily configured to facilitate access to video via the Internet from smartphones, laptops, and other mobile devices.

VIDEO STORAGE

A CCTV system needs to be designed and configured so that it retains the necessary quantity and quality of video data. CCTV systems must also be equipped with appropriate export and archiving capabilities.

Equally important is establishing the image resolution, image rate, and the number of days of recording that will be stored by the system. These factors will influence the use, access, recall, and storage requirements of a CCTV system

Types of video storage

1. Media Storage

Many organizations use write-once, read-many (WORM) media for long-term storage needs due to its secure and cost effective features. Current WORM technologies include optical discs such as CDs and DVDs, while older systems may use magnetic disks or tape.

A disadvantage of using WORM media is that record management can be cumbersome. For example, a CD can be destroyed, damaged, or easily removed from its environment, which is

not the case with data stored on servers. The storage capacity of WORM media is also a concern.

2.Scalable Network Storage

Data storage in CCTV systems is changing rapidly and has been influenced greatly by IP-based systems requiring efficient and cost effective storage. The market offers various network storage options for IP-based systems and hybrid systems, which incorporate both analog and IP technology and communications protocols.

Types of scalable network storage

1. Interface Protocols

Connecting servers to storage devices is typically accomplished through use of small computer systems interface (SCSI) protocol. Within CCTV systems, SCSI is the most dominantly used storage interface.

With the increase in available IP technologies and the need for interoperable and open architecture, the iSCSI protocol incorporates new capabilities to access voice, video and data from multiple types of network storage devices and make it available across an IP network.

2. Direct Attached Storage

DAS is considered an older technology that was developed as a stand-alone mechanism to connect hosts to storage devices through a direct, one-to-one SCSI attachment. Adding storage and servers to a DAS system to meet demands can result in a proliferation of server and storage islands.

In a DAS environment, storage sharing is limited because of its direct affiliation to the servers. DAS is still used today in CCTV systems, but external storage solutions are usually better options for CCTV video than fixed DAS storage. As LANs gained popularity, the server attached storage (SAS) was developed as an alternative to DAS in order to achieve a distributed approach via a LAN.

3. Storage Area Network

A SAN consists of communications infrastructure and management layers that ensure secure and robust data transfer. Storage appliances within the SAN contain data blocks. Operators access the data blocks when needed over the network. A SAN is a dedicated, high-performance network, typically using fiber channel technology,

Fiber channel protocol and interconnect technology provide high performance transfers of block data. SANs are generally used to connect numerous storage devices such as DVRs and NVRs to one or more centralized, shared storage systems.

4. Network Attached Storage

NAS systems record and access data in file format and consist of an engine that retrieves files from one or more storage devices. With NAS technology, servers maintain file systems on their local storage, and clients can access files at servers over a network via LAN or WAN technology, typically using Ethernet.

